

5. DriverManager class

The DriverManager class is the component of JDBC API and also a member of the *java.sql* package. The DriverManager class acts as an interface between users and drivers. It keeps track of the drivers that are available and handles establishing a connection between a database and the appropriate driver. It contains all the appropriate methods to register and deregister the database driver class and to create a connection between a Java application and the database. The DriverManager class maintains a list of Driver classes that have registered themselves by calling the method `DriverManager.registerDriver()`. Note that before interacting with a Database, it is a mandatory process to register the driver; otherwise, an exception is thrown.

Methods of the DriverManager Class

Method	Description
1) public static synchronized void <code>registerDriver(Driver driver)</code>	is used to register the given driver with DriverManager. No action is performed by the method when the given driver is already registered.
2) public static synchronized void <code>deregisterDriver(Driver driver)</code>	is used to deregister the given driver (drop the driver from the list) with DriverManager. If the given driver has been removed from the list, then no action is performed by the method.
3) public static Connection <code>getConnection(String url)</code> throws <code>SQLException</code>	is used to establish the connection with the specified url. The <code>SQLException</code> is thrown when the corresponding Driver class of the given database is not registered with the DriverManager.
4) public static Connection <code>getConnection(String url,String userName,String password)</code> throws <code>SQLException</code>	is used to establish the connection with the specified url, username, and password. The <code>SQLException</code> is thrown when the corresponding Driver class of the given database is not registered with the DriverManager.
5) public static Driver <code>getDriver(String url)</code>	Those drivers that understand the mentioned URL (present in the parameter of the method) are returned by this method provided those drivers are mentioned in the list of registered drivers.
6) public static int <code>getLoginTimeout()</code>	The duration of time a driver is allowed to wait in order to establish a connection with the database is returned by this method.

7) public static void setLoginTimeout(int sec)	The method provides the time in seconds. sec mentioned in the parameter is the maximum time that a driver is allowed to wait in order to establish a connection with the database. If 0 is passed in the parameter of this method, the driver will have to wait infinitely while trying to establish the connection with the database.
8) public static Connection getConnection (String URL, Properties prop) throws SQLException	A connection object is returned by this method after creating a connection to the database present at the mentioned URL, which is the first parameter of this method. The second parameter, which is "prop", fetches the authentication details of the database (username and password.). Similar to the other variation of the getConnection() method, this method also throws the SQLException, when the corresponding Driver class of the given database is not registered with the DriverManager.